# LLM & Agent Evaluation Framework

## Multi-Model Benchmarking, Cost Analysis, and Agent Trajectory Evaluation

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# Agenda


1. Project Overview
2. Quality Benchmarks
3. Operational Metrics
4. Safety & Calibration
5. Statistical Testing
6. Agent Evaluation
7. Conclusions

# Why LLM Evaluation Matters

## The Problem

Choosing among LLMs requires balancing many factors:

- Quality across diverse task types
- Latency for user-facing applications
- Cost at production scale
- Safety & hallucination rates
- Agent capabilities (tool use, planning)

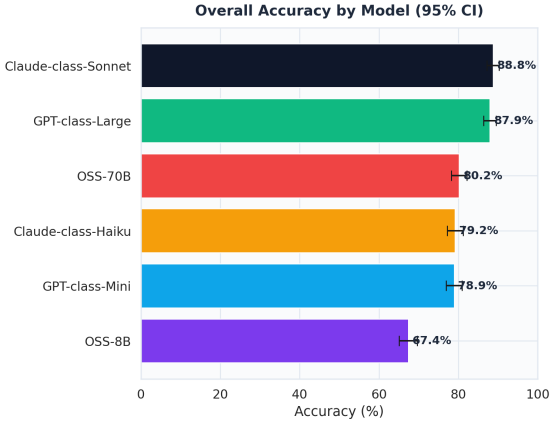
## Our Framework

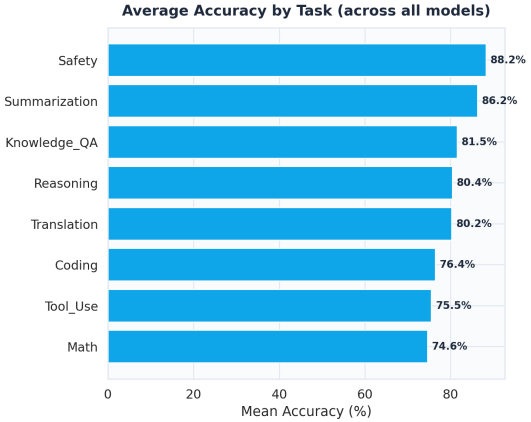
A reproducible evaluation suite covering all these dimensions, with statistical rigor and cost-aware decision support.

## Scope

| | |
|---|---|
| Models compared | 6 |
| Task categories | 8 |
| Questions per cell | 200 |
| **Total evaluations** | **9,600** |
| Agent trajectories | 500 |

# Overall Accuracy with Confidence Intervals

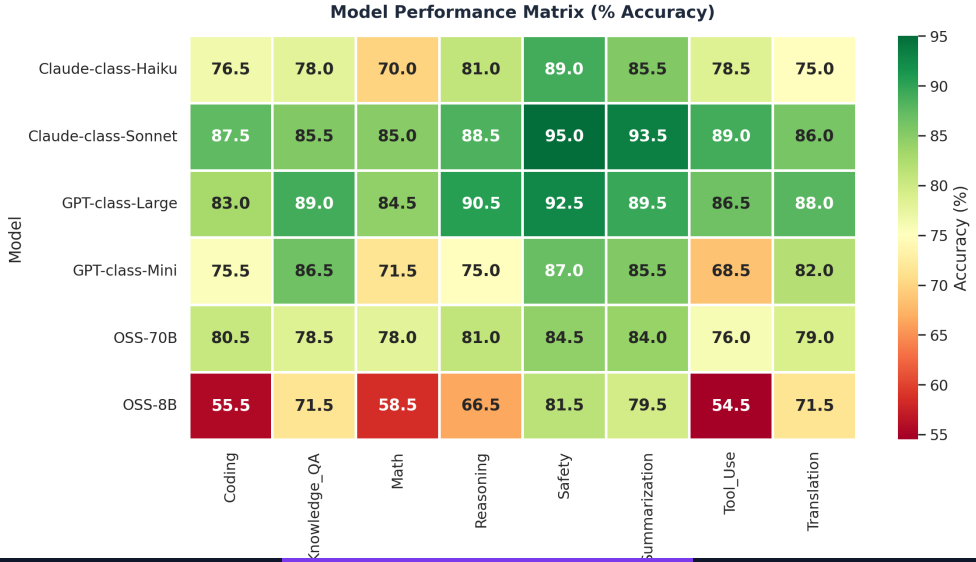


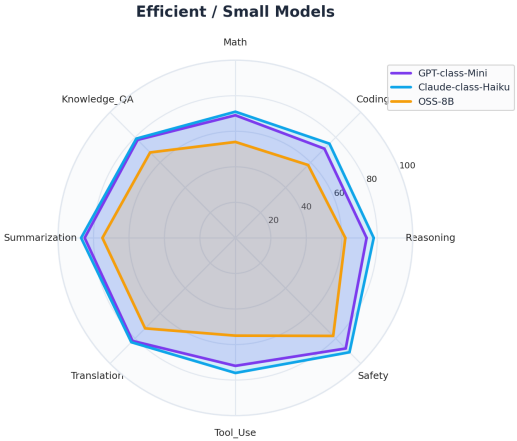

## Leaderboard

Claude-class-Sonnet leads at 88.8%, closely followed by GPT-class-Large (87.9%). The 95% confidence intervals are tight thanks to 1,600 evaluations per model.

# Performance Heatmap (Model × Task)



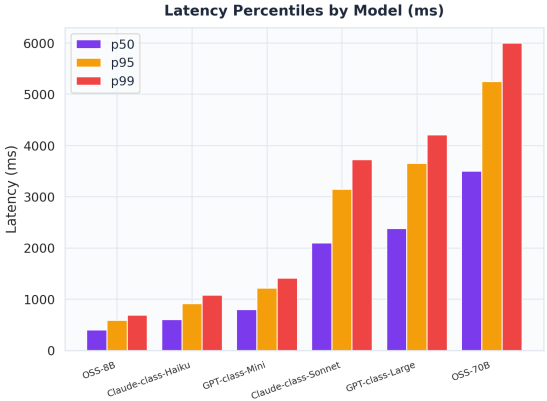
# Skill Profile Comparison

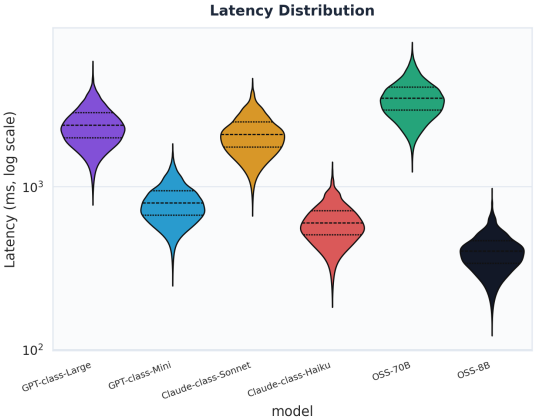



## Radar Charts

Side-by-side comparison of premium vs efficient model tiers across all 8 task dimensions

# Latency Analysis

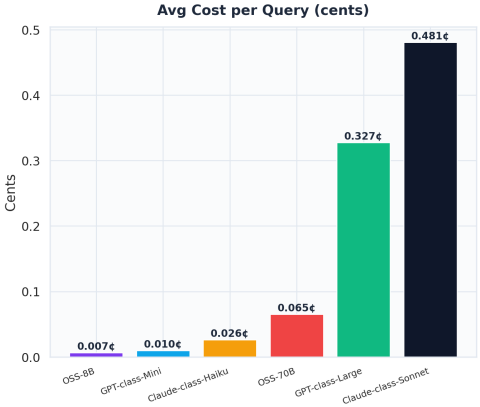


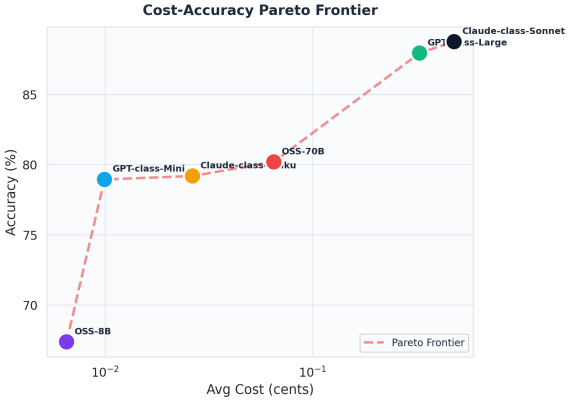




## Latency Spread

p50 latencies range from 400ms (OSS-8B) to 3.5s (OSS-70B). Tail latencies (p95/p99) matter for user experience; a 4–5x ratio of p99/p50 is typical.

# Cost-Accuracy Pareto Frontier

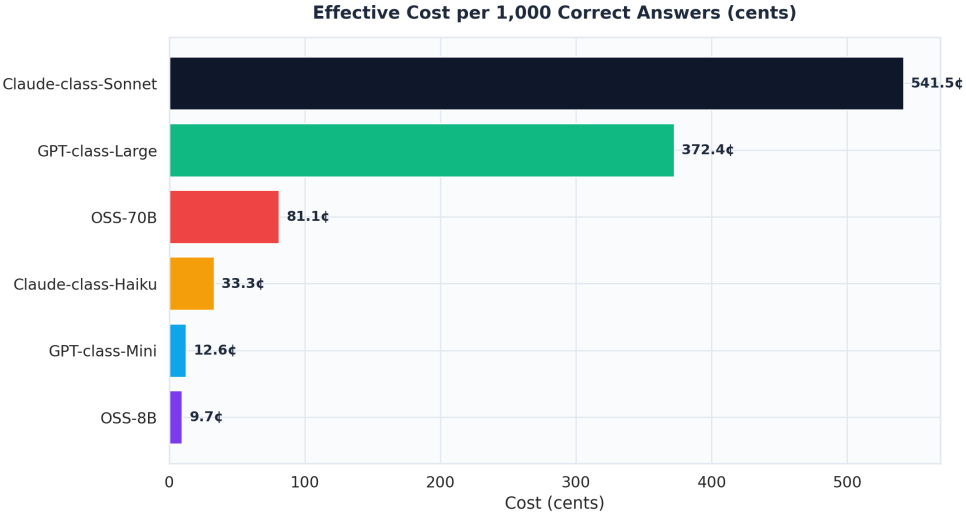


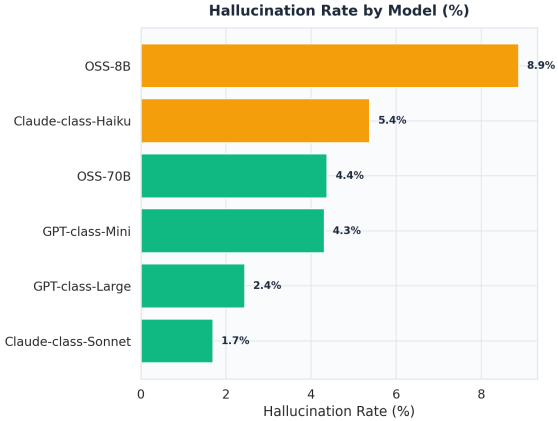

## The Frontier

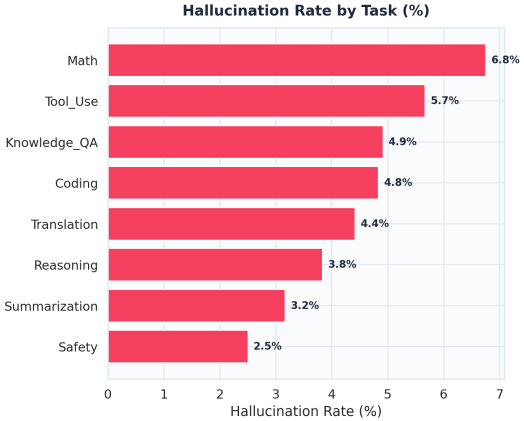
Models above the dashed line dominate; models below are Pareto-suboptimal. **OSS-8B**, **GPT-class-Mini**, **Claude-class-Haiku**, **GPT-class-Large**, and **Claude-class-Sonnet** all sit

# Effective Cost (per Correct Answer)



# Hallucination Rates

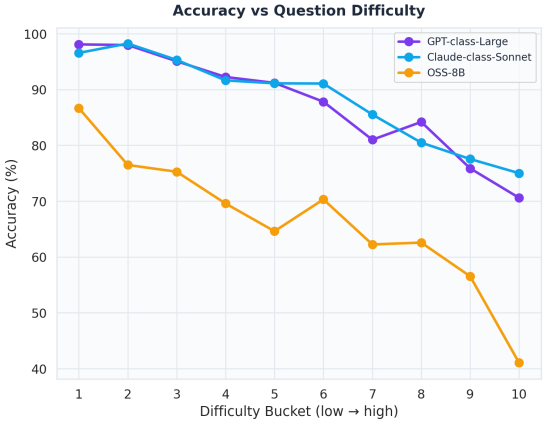


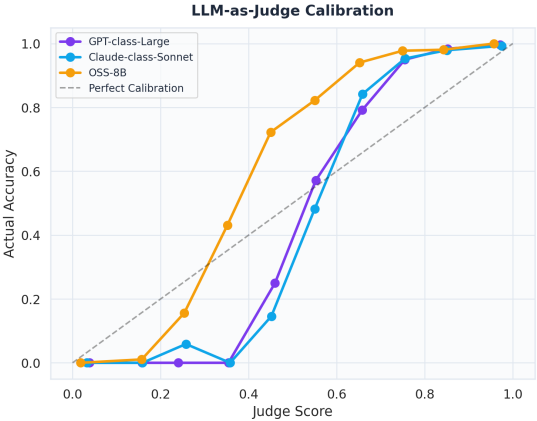




## Critical Insight

Hallucination is heavily task-dependent: math and reasoning trigger more hallucinations than translation or summarization. Smaller models hallucinate more on hard tasks.

# Difficulty Scaling & Judge Calibration

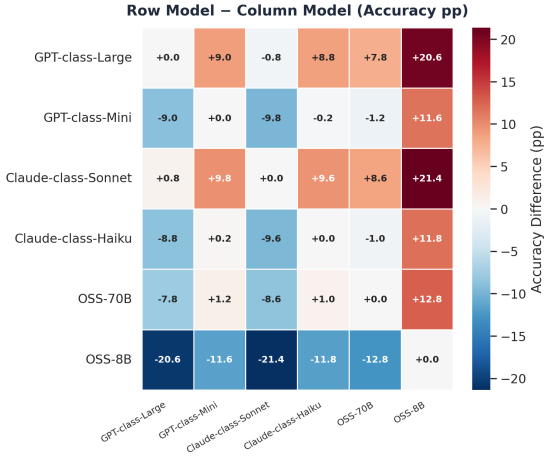


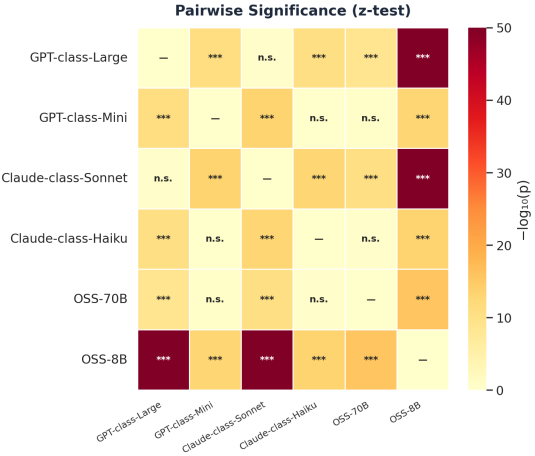




## LLM-as-Judge

The judge model produces calibrated confidence scores — predicted scores match observed accuracy. This validates LLM-as-judge as a reliable evaluation methodology.

# Pairwise Significance Testing

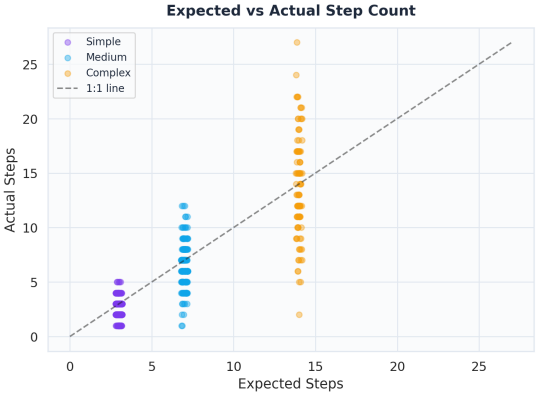



## Statistical Rigor

Two-proportion z-tests on every model pair. Top 2 models (Claude-Sonnet, GPT-Large) are

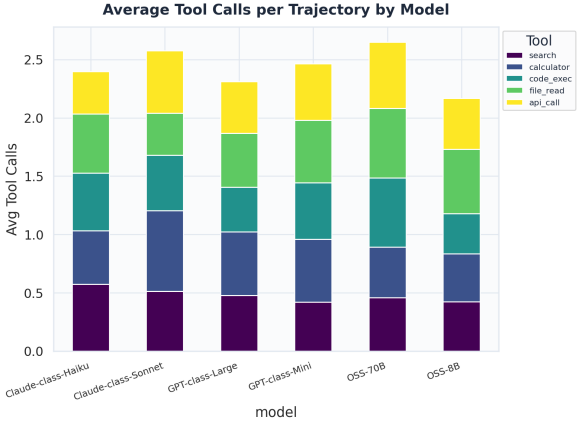
# Agent Success by Complexity




## Complexity Wall

All models drop sharply on complex tasks, but stronger models degrade more gracefully. The actual-vs-expected step count reveals planning quality.

# Tool Usage Patterns



## Operational Insights

Different models prefer different tools. Error and retry rates correlate inversely with success rate — weaker models error more and don't always recover.

# Executive Summary

## LLM Evaluation — Executive Summary

### Model Leaderboard (Ranked by Accuracy)

| | Accuracy % | p50 Latency (ms) | Avg Cost (cents) | Hallucination % |
|---|---|---|---|---|
| **Claude-class-Sonnet** | 88.8 | 2095.673 | 0.481 | 1.7 |
| **GPT-class-Large** | 87.9 | 2384.019 | 0.327 | 2.4 |
| **OSS-70B** | 80.2 | 3499.181 | 0.065 | 4.4 |
| **Claude-class-Haiku** | 79.2 | 601.237 | 0.026 | 5.4 |
| **GPT-class-Mini** | 78.9 | 796.697 | 0.01 | 4.3 |
| **OSS-8B** | 67.4 | 402.517 | 0.007 | 8.9 |

```
USE CASE RECOMMENDATIONS

 Best Overall Quality
    → Claude-class-Sonnet
    → 87.7% accuracy

⚡ Best Latency
    → OSS-8B
    → 423ms p50

 Best Cost-Efficiency
    → OSS-8B
    → 0.07¢/query

 Best Safety
    → Claude-class-Sonnet
    → 95% safety, 2.3% halluc

 Best for Agents
    → Claude-class-Sonnet
    → Highest tool-use score
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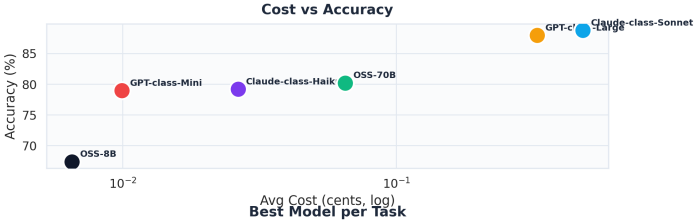

# Future Work & Production Integration

## Framework Capabilities

- Quality + Operational + Safety
- Statistical significance testing
- Agent trajectory analysis
- LLM-as-Judge calibration
- Pareto-optimal recommendations

## Future Extensions

1. Human preference data integration
2. Adversarial / red-team test suite
3. Context-length scaling experiments
4. Multi-turn conversation eval
5. RAG faithfulness metrics
6. Public benchmark integration
7. CI/CD regression testing

# Thank You

Questions & Discussion

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